Bureau
International des
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Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Indium 115 Ion ()

Indium 115 Ion ()



CCL-CCTF Frequency Standards Working Group

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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2: Indium 115 Ion ()

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Appendix 2

1. Recommended value [1] of the frequency in the CIPM List of Frequencies

$f\left(^{115}\mathrm{In}^{+}\right) = 1\ 267\ 402\ 452\ 901\ 050\ \mathrm{Hz}$

equivalent to

$\lambda\left(^{115}\mathrm{In}^{+}\right) = 236\ 540\ 853.549\ 544\ \mathrm{fm},$

with a relative standard uncertainty of 1.6×10^{-14}.

2. Source data

Adopted value $f\left(^{115}\mathrm{In}^{+}\right) = 1\ 267\ 402\ 452\ 901\ 050\ \mathrm{Hz}$

with a relative uncertainty of 1.6×10^{-14} calculated from

$f\left(^{115}\mathrm{In}^{+}\right)$/ Hz	u/ Hz	source data
1 267 402 452 899 920	690	[2] (*a*)
1 267 402 452 901 265	256	[3]
1 267 402 452 901 049.9	20.7	[4] (*b*)

(*a*) From the least square procedure it turned out that the value and the uncertainty given in [2] is not compatible with the remaining data. Thus the uncertainty given in the original publication [2] was increased to 690 Hz to make it statistically more consistent.

(*b*) The uncertainty of 6.9 Hz presented in ref. [4] was increased by a factor of three because it dominates the result with its uncertainty much lower than any of the others. Another fourth measurement [5] has not been used since the data were identical with the ones of ref. [3] but a much lower uncertainty of 63 Hz was reported without reasoning. The modification of the uncertainties followed the procedures outlined in ref. [6].

as a weighted mean of the above values.

$^{115}In^{+}$, $5s^2\ ^1S_0 - 5s5p\ ^3P_0$ unperturbed optical transition

References

[1] Consultative Committee for Time and Frequency (CCTF): Report of the 21st meeting (8-9 June 2017) to the International Committee for Weights and Measures; https://www.bipm.org/utils/common/pdf/CC/CCTF/CCTF21.pdf.

[2] J. von Zanthier, Th. Becker, M. Eichenseer, A. Yu. Nevsky, Ch. Schwedes, E. Peik E., H. Walther, R. Holzwarth, J. Reichert, Th. Udem, T. W. Hänsch, P. V. Pokasov, M. N. Skvortsov, S. N. Bagayev: Absolute frequency measurement of the In^+ clock transition with a mode-locked laser; *Opt. Lett.* **25** 1729 – 1731 (2000)

[3] Y. H. Wang, R. Dumke, T. Liu, A. Stejskal, Y. N. Zhao, J. Zhang, Z. H. Lu, L. J. Wang, T. Becker, H. Walther: Absolute frequency measurement and high resolution spectroscopy of $^{115}In^+$ $5s^2\ ^1S_0 - 5s5p\ ^3P_0$ narrowline transition; *Opt. Commun.* **273**, 526 – 531 (2007)

[4] N. Ohtsubo, Y. Li, K. Matsubara, T. Ido, K. Hayasaka: Frequency measurement of the clock transition of an indium ion sympathetically-cooled in a linear trap; Opt. Express **25** 11725 – 11735 (2017)

[5] Y. H. Wang, T. Liu, R. Dumke, A. Stejskal, Y. N. Zhao, J. Zhang, Z. H. Lu, L. J. Wang, T. Becker, H. Walther: Improved Absolute Frequency Measurement of the $^{115}In^+ 5s^2\ ^1S_0 - 5s5p\ ^3P_0$ Narrowline Transition: Progress Towards an Optical Frequency Standard; *Laser Physics* **17** 1017 – 1024 (2007)

[6] F. Riehle, P. Gill, F. Arias, L. Robertsson: The CIPM List of Recommended Frequency Standard Values: Guidelines and Procedures; *Metrologia* **55**, 188 – 200 (2018)